

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)			hydrogen is a highly reactive gas ☐			1	1		
				only 0.5 ppm of hydrogen is present ☐						
				hydrogen does not become liquid on cooling to -200°C ☒						
				hydrogen has a higher boiling point than helium ☐						
	(b)			carbon dioxide has a boiling point above -200°C ☐			1	1		
				carbon dioxide has a melting point above -200°C ☒						
				carbon dioxide has a melting point below -200°C ☐						
				carbon dioxide has a boiling point below -200°C ☐						
	(c)			they have different boiling points (1) nitrogen has lowest boiling point and evaporates first / oxygen has highest boiling point and evaporates last (1) gases collected (at different places on column) in order of boiling points (1)	1	2		3		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)			7.53×10^7 (2) if incorrect award (1) for either of following 75 268 817 $\frac{700000}{0.0093}$			2	2	2	
				Question 6 total	1	2	4	7	2	0